6.6 THE BUCK-BOOST CONVERTER

Another basic switched-mode converter is the buck-boost converter shown in Fig. 6-11. The output voltage of the buck-boost converter can be either higher or lower than the input voltage.

Voltage and Current Relationships

Assumptions made about the operation of the converter are as follows:

1. The circuit is operating in the steady state.
2. The inductor current is continuous.
3. The capacitor is large enough to assume a constant output voltage.
4. The switch is closed for time DT and open for $(1-D)T$.
5. The components are ideal.

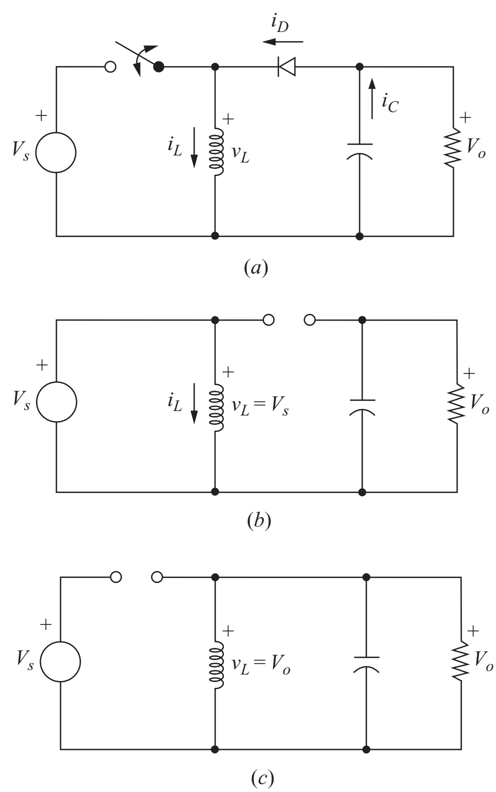


Figure 6-11 Buck-boost converter. (*a*) Circuit; (*b*) Equivalent circuit for the switch closed; (*c*) Equivalent circuit for the switch open.

Analysis for the Switch Closed When the switch is closed, the voltage across the inductor is

$$v_L = V_s = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_s}{L}$$

The rate of change of inductor current is a constant, indicating a linearly increasing inductor current. The preceding equation can be expressed as

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_s}{L}$$

Solving for Δi_L when the switch is closed gives

$$(\Delta i_L)_{\text{closed}} = \frac{V_s DT}{L} \tag{6-45}$$

Analysis for the Switch Open When the switch is open, the current in the inductor cannot change instantaneously, resulting in a forward-biased diode and current into the resistor and capacitor. In this condition, the voltage across the inductor is

$$v_L = V_o = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_o}{L}$$

Again, the rate of change of inductor current is constant, and the change in current is

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{(1-D)T} = \frac{V_o}{L}$$

Solving for Δi_L,

$$(\Delta i_L)_{\text{open}} = \frac{V_o(1-D)T}{L} \tag{6-46}$$

For steady-state operation, the net change in inductor current must be zero over one period. Using Eqs. (6-45) and (6-46),

$$(\Delta i_L)_{\text{closed}} + (\Delta i_L)_{\text{open}} = 0$$

$$\frac{V_s DT}{L} + \frac{V_o(1-D)T}{L} = 0$$

Solving for V_o,

$$\boxed{V_o = -V_s\left(\frac{D}{1-D}\right)} \tag{6-47}$$

The required duty ratio for specified input and output voltages can be expressed as

$$D = \frac{|V_o|}{V_s + |V_o|} \tag{6-48}$$

The average inductor voltage is zero for periodic operation, resulting in

$$V_L = V_s D + V_o(1 - D) = 0$$

Solving for V_o yields the same result as Eq. (6-47).

Equation (6-47) shows that the output voltage has opposite polarity from the source voltage. *Output voltage magnitude of the buck-boost converter can be less than that of the source or greater than the source, depending on the duty ratio of the switch.* If $D > 0.5$, the output voltage is larger than the input; and if $D < 0.5$, the output is smaller than the input. Therefore, this circuit combines the capabilities of the buck and boost converters. Polarity reversal on the output may be a disadvantage in some applications, however. Voltage and current waveforms are shown in Fig. 6-12.

Note that the source is never connected directly to the load in the buck-boost converter. Energy is stored in the inductor when the switch is closed and transferred to the load when the switch is open. Hence, the buck-boost converter is also referred to as an *indirect* converter.

Power absorbed by the load must be the same as that supplied by the source, where

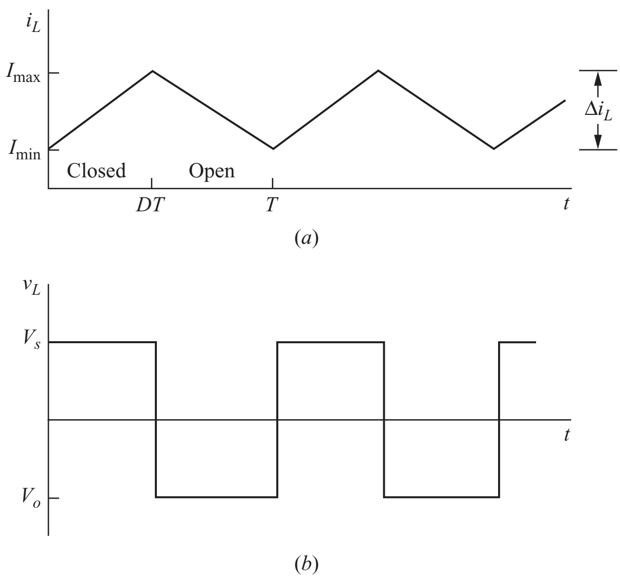
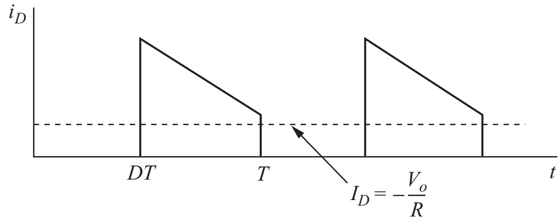
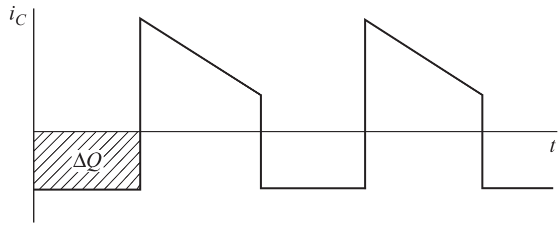


Figure 6-12 Buck-boost converter waveforms. (*a*) Inductor current; (*b*) Inductor voltage; (*c*) Diode current; (*d*) Capacitor current.



(c)



(d)

Figure 6-12 *(continued)*

$$P_o = \frac{V_o^2}{R}$$

$$P_s = V_s I_s$$

$$\frac{V_o^2}{R} = V_s I_s$$

Average source current is related to average inductor current by

$$I_s = I_L D$$

resulting in

$$\frac{V_o^2}{R} = V_s I_L D$$

Substituting for V_o using Eq. (6-47) and solving for I_L, we find

$$I_L = \frac{V_o^2}{V_s R D} = \frac{P_o}{V_s D} = \frac{V_s D}{R(1-D)^2} \tag{6-49}$$

Maximum and minimum inductor currents are determined using Eqs. (6-45) and (6-49).

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = \frac{V_s D}{R(1-D)^2} + \frac{V_s DT}{2L} \tag{6-50}$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = \frac{V_s D}{R(1-D)^2} - \frac{V_s DT}{2L} \tag{6-51}$$

For continuous current, the inductor current must remain positive. To determine the boundary between continuous and discontinuous current, $I_{\min}$ is set to zero in Eq. (6-51), resulting in

$$(Lf)_{\min} = \frac{(1 - D)^2 R}{2} \tag{6-52}$$

or

$$\boxed{L_{\min} = \frac{(1 - D)^2 R}{2f}} \tag{6-53}$$

where f is the switching frequency.

Output Voltage Ripple

The output voltage ripple for the buck-boost converter is computed from the capacitor current waveform of Fig. 6-12*d*.

$$|\Delta Q| = \left(\frac{V_o}{R}\right) DT = C \Delta V_o$$

Solving for ΔV_o,

$$\Delta V_o = \frac{V_o DT}{RC} = \frac{V_o D}{RCf}$$

or

$$\boxed{\frac{\Delta V_o}{V_o} = \frac{D}{RCf}} \tag{6-54}$$

As is the case with other converters, the equivalent series resistance of the capacitor can contribute significantly to the output ripple voltage. The peak-to-peak variation in capacitor current is the same as the maximum inductor current. Using the capacitor model shown in Fig. 6-6, where $I_{L,\max}$ is determined from Eq. (6-50),

$$\Delta V_{o,\mathrm{ESR}} = \Delta i_C r_C = I_{L,\max} r_C \tag{6-55}$$

EXAMPLE 6-6

Buck-Boost Converter

The buck-boost circuit of Fig. 6-11 has these parameters:

$V_s = 24$ V
$D = 0.4$
$R = 5\ \Omega$
$L = 20\ \mu$H
$C = 80\ \mu$F
$f = 100$ kHz

Determine the output voltage, inductor current average, maximum and minimum values, and the output voltage ripple.

■ **Solution**

Output voltage is determined from Eq. (6-47).

$$V_o = -V_s\left(\frac{D}{1-D}\right) = -24\left(\frac{0.4}{1-0.4}\right) = -16 \text{ V}$$

Inductor current is described by Eqs. (6-49) to (6-51).

$$I_L = \frac{V_s D}{R(1-D)^2} = \frac{24(0.4)}{5(1-0.4)^2} = 5.33 \text{ A}$$

$$\Delta i_L = \frac{V_s DT}{L} = \frac{24(0.4)}{20(10)^{-6}(100{,}000)} = 4.8 \text{ A}$$

$$I_{L,\max} = I_L + \frac{\Delta i_L}{2} = 5.33 + \frac{4.8}{2} = 7.33 \text{ A}$$

$$I_{L,\min} = I_L - \frac{\Delta i_L}{2} = 5.33 - \frac{4.8}{2} = 2.93 \text{ A}$$

Continuous current is verified by $I_{\min} > 0$.
Output voltage ripple is determined from Eq. (6-54).

$$\frac{\Delta V_o}{V_o} = \frac{D}{RCf} = \frac{0.4}{(5)(80)(10)^{-6}(100{,}000)} = 0.01 = 1\%$$

6.7 THE ĆUK CONVERTER

The Ćuk switching topology is shown in Fig. 6-13*a*. Output voltage magnitude can be either larger or smaller than that of the input, and there is a polarity reversal on the output.

The inductor on the input acts as a filter for the dc supply to prevent large harmonic content. Unlike the previous converter topologies where energy transfer is associated with the inductor, energy transfer for the Ćuk converter depends on the capacitor C_1.

The analysis begins with these assumptions:

1. Both inductors are very large and the currents in them are constant.
2. Both capacitors are very large and the voltages across them are constant.
3. The circuit is operating in steady state, meaning that voltage and current waveforms are periodic.
4. For a duty ratio of D, the switch is closed for time DT and open for $(1-D)T$.
5. The switch and the diode are ideal.